



DPUSGEWorld

Manages and updates a large number of graphical objects (altogether called a 'world').

1. Dynamic parameters:

SaveShaderInfo

Saves information about all loaded shaders to the folder `\SILAB\Data\shader`.

2. Static parameters:

RenderMode

Indicates which special effects are active. Possible values are 0..8. A detailed description follows in the section "RenderMode".

Default: 4

AntiAlias

Amount of anti-aliasing (smoothing of object borders). Possible values are (depending on the graphics hardware): 0 = off, 2..16 = enabled

Using this parameter, anti-aliasing is set for all windows created by instances of *DPUSGEView*.

Note: If the render modes "Nighttime" or "Nighttime with Shadows" are selected, anti-aliasing reduces performance even more than usual.

Default: 8

PreferMultisampling

Selects a method to perform anti-aliasing in the "Nighttime" render modes (*RenderMode* = 5 or 6).

If set to *true* (the default), multisampling is used if supported by the hardware. If set to *false*, supersampling is used if *AntiAlias* $\geq$ 8, and otherwise an emulation.

ShadowQuality

Sets the resolution of shadow maps. Higher settings can reduce the frame rate, but lead to smoother shadow borders. Possible values are 1..5.

If values greater than 3 are used, it is recommended to also increase the *MaxShadowDist* parameter of *DPUSGELightParams*.

Default: 4

ReflectionQuality

Sets the resolution of the reflection view. Higher settings are more expensive, but the reflected image's quality is improved. Possible values are 1..3.

Default: 2

AlwaysUpdateShadow

Boolean flag which selects if all shadow buffers are updated in each trigger step (*true*) or if they are updated alternately (*false*).

Activating the flag is more expensive, but can reduce the flickering of shadows, especially of quickly moving vehicles on the road.

Default: *false*



<i>SupportMoreImageFormats</i>	Activates support for additional image formats, which can be used in textures. The default value is <i>false</i>, meaning that only the lossless file formats PNG and TGA are supported. All textures that are included in SILAB's scope of delivery use one of these file formats. If set to <i>true</i>, the image formats JPG, BMP, GIF and ICO are also supported.
<i>SortTransparentObjects</i>	If set to <i>true</i> (the default value), transparent objects are sorted by their distance to the first view. This usually improves display quality. If the algorithm doesn't work correctly (for example, when multiple views with different viewing directions are rendered), it can be disabled by setting the value to <i>false</i>.
<i>HeadlightLowerTex</i>	Resource name of the texture used to model the light distribution of a headlamp's lower beam (in render mode 2, 4, or greater). The default texture <code>headlights.default_lower_90</code> is suitable for cars and trucks.
<i>HeadlightUpperTex</i>	Resource name of the texture used to model the light distribution of a headlamp's upper beam (in render mode 2, 4, or greater). The default texture <code>headlights.default_upper_90</code> is suitable for cars and trucks.

3. Dependencies to other DPUs:

If **DPUSGEWorld** runs together with **DPUSGEView**, the *Index* value of **DPUSGEWorld** must be less than the *Index* of **DPUSGEView**.

4. Functional principle:

A SGE world represents a large amount of graphics objects which are managed by **DPUSGEWorld** and displayed in a window created by **DPUSGEView**.

The objects visible within the SGE world are inserted and animated by other DPUs, e.g. **DPUSCNXSGE**, **DPUTRFSGE** and **DPUSGEObject**.

5. Render mode:

The following render modes (static parameter *RenderMode*) are implemented in the visualization SGE. As a guideline, the higher the *RenderMode*, the higher the requirements for graphics board and processor.

Index	Name	Description
0	No Lighting	The scene is not lit at all. The result is similar to SILAB's previous visualization SIG.
1	Daytime	Surfaces are shaded according to the direction of the sun. However, no shadows are visible.
2	Daytime with Headlight	Like Daytime, but additionally considers the the EGO vehicle's headlights.
3	Daytime with Shadows	Like Daytime, but additionally casts shadows for the "Sun" light source.



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4	Daytime with Shadows and Headlight	Like Daytime with Shadows, but the scene is additionally lit by the vehicle's headlights. However, the headlights do not cast shadows.
5	Nighttime	Surfaces are lit by the sun, by streetlights and by the EGO and other vehicle's headlights. However, no shadows are visible.
6	Nighttime with Shadows	Like Nighttime, but additionally, the "Sun" light source casts shadows. (All other light sources do not.)
7	Daytime with Headlight and Reflection	Like Daytime with Headlight, but a view from below the road is additionally computed, which is used to display realistic reflections on the road surface (e.g. when it is wet). Note: If no reflecting surface is visible at a certain point of the road network, updates for the reflection view can be disabled by setting the variable <i>ReflectionVisible</i> of <i>DPUEnvControl</i> to 0. This increases the display frame rate and thus may remove stuttering.
8	Daytime with Headlight, Shadows and Reflection	Like Daytime with Headlight and Reflection, but additionally, the "Sun" light source casts shadows.